EXAMINATION

Luleå University of Technology

Course: ***D7024E***
Course name: ***Distributed and Mobile Computing Systems***
Date: ***2025-11-19 to 2025-12-09***
Time:
Aid: ***WWW and Generative AI (e.g., ChatGPT, Bard) (no interactive communication with other people)***

Teacher on duty (complete telephone number): Johan Kristiansson, 070-3632912 johan.kristiansson@ltu.se	***Teacher on duty (complete telephone number):*** Olov Schelén, 0920 492237 olov.schelen@ltu.se
Teacher on duty (complete telephone number):	***Teacher on duty (complete telephone number):***

Grade scale: ***Fail (U) or Pass (g)***	Note that this grade is for the exam. The exam is supposed to be easy enough so that anyone that followed the course properly should pass it. The grade for the course (including all assessment criteria) may be different from the grade of the exam.
Total number of questions and score: ***total score 1***	
Other information: This is an individual home exam.	Note: No interactive cooperation with other people is allowed. Information can be searched on the Web and with generative AI (e.g., ChatGPT, Bard, etc.). Please provide references when such information is used in your answers.

General instructions

You shall formulate answers with your own words; if you use generative AI a personal reflection must be included. In any case reference your primary sources (e.g., URL). Your ***solution must be uploaded in the Canvas D7024E room under Assignments***, at the latest at 17:00 (unless you have gotten approval for extra time). ***Please put your name both in the filename and on the first line of your document.***

After examination

Examination results are posted on Canvas within 4 weeks after the examination.

Extra Exam Question

GenAI tools can speed up the development of distributed systems, including your implementation of the Kademlia DHT protocol. At the same time, automatically generated code raises questions about whether the system actually behaves correctly, especially when it runs with many nodes and must handle continuously changing network conditions.

Write an extended essay of at least 3 pages (excluding diagrams and pseudo code) discussing how you would build confidence in the correctness of code generated with the help of GenAI in the context of a Kademlia-based distributed hash table. Your answer should explore what correctness even means for such a system, how you would attempt to validate it, and what kinds of reasoning or evidence you would consider convincing. Address issues such as node lookup, routing behaviour, and interactions between large numbers of nodes.

As part of your essay, include a short pseudo code example that illustrates how you might test or assert the correctness of a critical aspect of your Kademlia implementation when operating at scale. Use any level of abstraction you feel appropriate. The point is to show how you would think about checking correctness, not to present a full test suite.

You may structure your essay however you like. Focus on explaining your reasoning clearly, discussing the kinds of failures or inconsistencies you would worry about, the forms of validation you consider meaningful, and the challenges introduced when GenAI contributes to the codebase. Conclude by reflecting on what kinds of uncertainty remain even after extensive validation.